MATRICES

A matrix is a rectangular array of items or numbers. These items or numbers are arranged in rows and columns to represent some information.

The position of an element in one matrix is very important as well be seen later; therefore an element is located by the number of the row and column which it occupies.

The size of a matrix is defined by the number of its rows (m) and column (n).

For example = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and B = $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

are (2 x 2) and (3 x 3) matrices since A has 2 rows and 2 columns and B has 3 rows and 3 columns.

A matrix A with three rows and four columns is given by one of:

$$A=\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix}$$

or

$$A = (a_{ij}) \quad i = 1, 2, 3$$

$$j = 1, 2, 3, 4$$ *where i represents the row number whereas j represents the column number*

Properties of matrices

Equal Matrices

Two matrices A and B are said to be equal, that is

$$A = B \qquad \text{or} \quad (a_{ij}) = (b_{ij})$$

If and only if they are identical if they both have the same number of rows and columns and the elements in the corresponding locations in the two matrices should be the same, that is, $a_{ij} = b_{ij}$ for all i. And j.

Example

The following matrices are equal $\begin{pmatrix} 3 & 4 & 0 \\ 2 & 2 & 3 \\ 5 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 4 & 0 \\ 2 & 2 & 3 \\ 5 & 1 & 1 \end{pmatrix}$

Column Matrix or column vector

A column matrix, also referred to as column vector is a matrix consisting of a single column.

For example $x = \begin{pmatrix} x_1 \\ x_2 \\ . \\ . \\ . \\ x_n \end{pmatrix}$

Row matrix or row vector

It is a matrix with a single row

For example $y = (y_1, y_2, y_3 \ldots\ldots\ldots y_n)$

Transpose of a Matrix

The transpose of an mxn matrix A is the nxm matrix A^T obtained by interchanging the rows and columns of A.

$$A = \left[a_{ij} \right]$$

The transpose of A i.e. A^T is given by

$$A^T = \left[a_{ij} \right]_{mxn} = \left[a_{ji} \right]_{nxm}$$

Example

Find the transposes of the following matrices

$$A = \begin{pmatrix} 1 & 5 & 7 \\ 2 & 1 & 4 \\ 0 & 9 & 3 \end{pmatrix}$$

$$B = (b_1, b_2, b_3, b_4)$$

$$C = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Solution

$$\text{i. } A^T = \begin{pmatrix} 1 & 5 & 7 \\ 2 & 1 & 4 \\ 0 & 9 & 3 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 & 0 \\ 5 & 1 & 9 \\ 7 & 4 & 3 \end{pmatrix}$$

$$\text{ii. } B^T = (b_1, b_2, b_3, b_4)^T = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$$

$$\text{iii. } C^T = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}^T = (x_1 \ x_2 \ x_3)$$

Square Matrix

A matrix A is said to be square when it has the same number of rows as columns

e.g.

$$A = \begin{pmatrix} 2 & 5 \\ 3 & 7 \end{pmatrix} \quad \text{is a square matrix of order 2}$$

B = n × n is a square matrix of the order n

Diagonal matrices

It is a square matrix with zeros everywhere in the matrix except on the principal diagonal

e.g.

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 7 \end{pmatrix} \quad , \quad B = \begin{pmatrix} 9 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

An identity of unity matrix

It is a diagonal matrix in which each of the diagonal elements is a positive one (1)

e.g.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and } I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2×2 unit matrix $\qquad$ 3×3 unit matrix

A null or zero matrix

A null or zero matrix is a matrix whose elements are all equal to zero.

Sub matrix

The sub matrix of the matrix A is another matrix obtained from A by deleting selected row(s) and/or column(s) of the matrix A.

$$\text{e.g, if } A = \begin{pmatrix} 7 & 9 & 8 \\ 2 & 3 & 6 \\ 1 & 5 & 0 \end{pmatrix}$$

$$\text{then } A_1 = \begin{pmatrix} 2 & 3 & 6 \\ 1 & 5 & 0 \end{pmatrix} \text{ and } A_2 = \begin{pmatrix} 7 & 9 \\ 1 & 5 \end{pmatrix}$$

are both sub matrices of A

OPERATION ON MATRICES

Matrix addition and subtraction

We can add any number of matrices (or subtract one matrix from another) if they have the same sizes. Addition is carried out by adding together corresponding elements in the matrices. Similarly subtraction is carried out by subtracting the corresponding elements of two matrices as shown in the following example

Example: Given A and B, calculate A + B and A – B

$$A = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & -13 & -6 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 12 & 4 & -7 & 3 \\ 0 & -4 & 10 & -4 \\ 7 & -3 & 7 & 9 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & -13 & -6 & 0 \end{pmatrix} + \begin{pmatrix} 12 & 4 & -7 & 3 \\ 0 & -4 & 10 & -4 \\ 7 & -3 & 7 & 9 \end{pmatrix} = \begin{pmatrix} 18 & 3 & 3 & 8 \\ 3 & 0 & 12 & -9 \\ -2 & -16 & 1 & 9 \end{pmatrix}$$

$$A - B = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & -13 & -6 & 0 \end{pmatrix} - \begin{pmatrix} 12 & 4 & -7 & 3 \\ 0 & -4 & 10 & -4 \\ 7 & -3 & 7 & 9 \end{pmatrix} = \begin{pmatrix} -6 & -5 & 17 & 2 \\ 3 & 8 & -8 & -1 \\ -16 & -10 & -13 & -9 \end{pmatrix}$$

If it is assumed that A, B, C are of the same order, the following properties are fulfilled:

a) Commutative law: $A + B = B + A$

b) Associative law: $(A + B) + C = A + (B + C) = A + B + C$

Multiplying a matrix by a number

In this case each element of the matrix is multiplied by that number

Example

$$\text{If } A = \begin{pmatrix} 6 & -1 & 10 & 5 \\ 3 & 4 & 2 & -5 \\ -9 & 13 & -6 & 0 \end{pmatrix}$$

$$\text{then } (10)A = \begin{pmatrix} 60 & -10 & 100 & 50 \\ 30 & 40 & 20 & -50 \\ -90 & 130 & -60 & 0 \end{pmatrix}$$

Matrix Multiplication

a) **Multiplication of two vectors**

Let row vector A represent the selling price in shillings of one unit of commodity P, Q, R respectively and let column vector B represent the number of units of commodities P, Q, R sold respectively. Then the vector product A × B will be equal to the total sales value

i. e. $A \times B =$ Total sales value

$$\text{Let } A = \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} \text{ and } B = \begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix}$$

$$\text{then } \begin{pmatrix} 4 & 5 & 6 \end{pmatrix}\begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix} = 400 + 1{,}000 + 1{,}800 = \text{Shs } 3{,}200$$

Rules of multiplication

The row vector must have the same number of elements as the column vector

The first vector is a row vector and the second is a column vector

The corresponding elements in each vector are multiplied together and the results obtained are added. This addition is always a single number

Going back to the example given before

$$A \times B = \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix} = 4 \times 100 + 5 \times 200 + 6 \times 300 = \text{Shs}3{,}200, \text{ a single number}$$

b) Multiplication of two matrices

Rules

i. Multiplication is only possible if the first matrix has the same number of columns as the second matrix has rows. That is if A is the order a×b, then B has to be of the order b×c. If the A×B = D, then D must be of the order a×c.

ii. The general method of multiplication is that the elements in row m of the first matrix are multiplied by the corresponding elements in columns n of the second matrix and the products obtained are then added giving a single number.

We can express this rule as follows

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad \text{and } b = \begin{pmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{pmatrix}$$

$$\text{Then } A \times B = D = \begin{pmatrix} d_{11} & d_{12} & d_{13} \\ d_{21} & d_{22} & d_{23} \end{pmatrix}$$

A = 2 x 2 matrix B = 2 x 3 matrix D = 2 x 3 matrix

Where

$d_{11} = a_{11} \times b_{11} + a_{12} \times b_{21}$

$d_{12} = a_{11} \times b_{12} + a_{12} \times b_{22}$

Example I

$$\begin{pmatrix} 6 & 1 \\ 2 & 3 \end{pmatrix} \times \begin{pmatrix} 3 & 0 & 2 \\ 4 & 5 & 8 \end{pmatrix} = \begin{pmatrix} 6\times3+1\times4 & 6\times0+1\times5 & 6\times2+1\times8 \\ 2\times3+3\times4 & 2\times0+3\times5 & 2\times2+3\times8 \end{pmatrix}$$

$$= \begin{pmatrix} 22 & 5 & 20 \\ 18 & 15 & 28 \end{pmatrix}$$

Example II

Matrix X gives the details of component parts used in the make up of two products P_1 and P_2 matrix Y gives details of products made on each day of the week as follows:

$$\begin{array}{ccc} & & \text{Matrix X} \\ & & \text{Parts} \\ & & \begin{array}{ccc} A & B & C \end{array} \\ \text{Products} & \begin{array}{c} P_1 \\ P_2 \end{array} & \begin{bmatrix} 3 & 4 & 2 \\ 2 & 5 & 3 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} & \text{Matrix Y} \\ & \text{Products} \\ & \begin{array}{cc} P_1 & P_2 \end{array} \\ \begin{array}{c} \text{Mon} \\ \text{Tues} \\ \text{Wed} \\ \text{Thur} \\ \text{Fri} \end{array} & \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} \end{array}$$

Use matrix multiplication to find the number of component parts used on each day of the week.

Solution:

After careful consideration, it will be easy to decide that the correct order of multiplication is YXX (Note the order of multiplication). This multiplication is compatible and also it gives the desired answer.

$$Y \times X = \begin{pmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 2 \\ 2 & 2 \\ 1 & 1 \end{pmatrix} \times \begin{pmatrix} 3 & 4 & 2 \\ 2 & 5 & 3 \end{pmatrix} = \begin{pmatrix} 1\times3+2\times2 & 1\times4+2\times5 & 1\times2+2\times5 \\ 2\times3+3\times2 & 2\times4+3\times5 & 2\times2+3\times3 \\ 3\times3+2\times2 & 3\times4+2\times5 & 3\times2+2\times3 \\ 2\times3+2\times2 & 2\times4+2\times5 & 2\times2+2\times3 \\ 1\times3+1\times2 & 1\times4+1\times5 & 1\times2+1\times3 \end{pmatrix}$$

5 x 2 matrix   2 x 3 matrix   =   5 x 3 matrix

$$\begin{array}{cc} & \begin{array}{ccc} A & B & C \end{array} \\ \begin{array}{c} \text{Mon} \\ \text{Tues} \\ \text{Wed} \\ \text{Thur} \\ \text{Fri} \end{array} & \begin{pmatrix} 7 & 14 & 8 \\ 12 & 23 & 13 \\ 13 & 22 & 12 \\ 10 & 18 & 10 \\ 5 & 9 & 5 \end{pmatrix} \end{array}$$

Interpretation

On Monday, number of component parts A used is 7, B is 14 and C is 8. in the same way, the number of component parts used for other days can be interpreted.

The determinant of a square matrix

The determinant of a square matrix det (A) or $|A|$ is a number associated to that matrix. If the determinant of a matrix is equal to zero, the matrix is called singular matrix otherwise it is called non-singular matrix. The determinant of a non square matrix is not defined.

Determination of a 2 x 2 matrix

$$(A) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - cb$$

ii. Determinant of a 3 x 3 matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a\begin{pmatrix} e & f \\ h & i \end{pmatrix} - b\begin{pmatrix} d & f \\ g & i \end{pmatrix} + c\begin{pmatrix} d & e \\ g & h \end{pmatrix}$$

$$a\,(ei - fh) - b(di - gf) + c(dh - eg)$$

simplify

iii. Determinant of a 4 x 4 matrix

$$A = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$$

$$A = a\begin{pmatrix} f & g & h \\ j & k & l \\ n & o & p \end{pmatrix} - b\begin{pmatrix} e & g & h \\ i & k & l \\ n & o & p \end{pmatrix} + c\begin{pmatrix} e & f & h \\ i & j & l \\ m & o & p \end{pmatrix} - d\begin{pmatrix} e & f & g \\ i & j & k \\ m & n & o \end{pmatrix}$$

Simplify 3 x 3 determinants as in ii and then evaluate the 4 x 4 determinants.

Inverse of a matrix

If for an n (n square matrix A, there is another n (n square matrix B such that there product is the identity of the order n X n, In, that is A X B = B X A = I, then B is said to be inverse of A. Inverse if generally written as A-1

Hence AA-1 = I

Note: Only non singular matrices have an inverse and therefore the inverse of a singular matrix is non defined.

General method for finding inverse of a matrix

In order to introduce the rule to calculate the determinant as well as the inverse of a matrix, we should introduce the concept of minor and cofactor.

The minor of an element

Given a matrix A = (aij), the minor of an element aij in row i and column j (call it mij), is the value of the determinant formed by deleting row i and column j in matrix A.

Example

Let A = ‼ EMBED Equation.DSMT4 ¶ ⊥

The minors are,

$$m_{11} = \begin{vmatrix} 6 & 1 \\ 3 & 0 \end{vmatrix} = 6\times0 - 3\times1 = -3$$

$$m_{12} = \begin{vmatrix} 5 & 1 \\ 2 & 0 \end{vmatrix} = 5\times0 - 1\times2 = -2$$

Similarly

$$m_{13} = \begin{vmatrix} 5 & 6 \\ 2 & 3 \end{vmatrix} \quad m_{21} = \begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix} \quad m_{22} = \begin{vmatrix} 4 & 3 \\ 2 & 0 \end{vmatrix} \quad m_{23} = \begin{vmatrix} 4 & 2 \\ 2 & 3 \end{vmatrix}$$

$$= 15 - 12 = 3 \quad = 0 - 9 = -9 \quad = 0 - 6 = -6 \quad = 12 - 4 = 8$$

$$m_{31} = \begin{vmatrix} 2 & 3 \\ 6 & 1 \end{vmatrix} \quad m_{32} = \begin{vmatrix} 4 & 3 \\ 5 & 1 \end{vmatrix} \quad m_{33} = \begin{vmatrix} 4 & 2 \\ 5 & 6 \end{vmatrix}$$

$$= 2 - 18 = -16 \quad = 4 - 15 = -11 \quad = 24 - 10 = 14$$

The cofactor of an element

The cofactor of any element a_{ij} (known as c_{ij}) is the signed minor associated with that element.

The sign is not changed if (i+j) is even and it is changed if (i+j)is odd. Thus the sign alternated whether vertically or horizontally, beginning with a plus in the upper left hand corner.

i.e. 3 x 3 signed matrix will have signs $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$

Hence the cofactor of element a_{11} is m_{11} = -3, cofactor of a_{12} is $-m_{12}$ = +2 the cofactor of element a_{13} is $+m_{13}$ = 3 and so on.

$$\text{Matrix of cofactors of A} \quad = \quad \begin{pmatrix} -3 & 2 & 3 \\ 9 & -6 & -8 \\ -16 & 11 & 14 \end{pmatrix}$$

$$\text{in general for a matrix M} \quad = \quad \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

Cofactor of a is written as A, cofactor of b is written as B and so on.
Hence matrix of cofactors of M is written as

$$= \begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}$$

The determinant of a n×n matrix

The determinant of a n×n matrix can be calculated by adding the products of the element in any row (or column) multiplied by their cofactors. If we use the symbol Δ for determinant.

Then $\Delta = aA + bB + cC$

or

$= dD + eE + fF$ e.t.c

Note: Usually for calculation purposes we take $\Delta = aA + bB + cC$

Hence in the example under discussion

$$\Delta = (4 \times -3) + (2 \times 2) + (3 \times 3) = 1$$

The ad joint of a matrix

The ad joint of matrix $\begin{pmatrix} A & B & C \\ D & E & F \\ G & H & I \end{pmatrix}$ is written as

$\begin{pmatrix} A & D & G \\ B & E & H \\ C & F & I \end{pmatrix}$ i.e. change rows into columns and columns into rows (transpose)

The inverse of the matrix $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

is written as $\frac{1}{\text{determinant}} \times (\text{adjoint of the matrix of cofactors})$

i.e. $A^{-1} = \frac{1}{\Delta} \times \begin{pmatrix} A & D & G \\ B & E & H \\ C & F & I \end{pmatrix}$

Where Δ = aA + bB + cC

Hence inverse of $\begin{pmatrix} 4 & 2 & 3 \\ 5 & 6 & 1 \\ 2 & 3 & 0 \end{pmatrix}$

is found as follows

Δ = (4 × –3) + (2 × 2) + (3 (3) = 1

A = -3 B = 2 C = 3

D = 9 E = -6 F = -8

G = -16 H = 11I = 14

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(note: Check if A (A-1 = $A^{-1} \times A = 1$)

Solution of simultaneous equations

In order to determine the solutions of simultaneous equations, we may use either of the following 2 methods

i. The cofactor method
ii. Cramers rule

The cofactor method

This method requires that we obtain

a) The minors and cofactors
b) The adjoint of the matrix
c) The inverse of the matrix
d) Multiply the original by the inverse on both sides of the matrix equation

Example

Solve the following

a) $4x_1 + x_2 - 5x_3 = 8$
$-2x_1 + 3x_2 + x_3 = 12$
$3x_1 - x_2 + 4x_3 = 5$

b) $4x_1 + 3x_3 + 5x_3 = 27$
$x_1 + 6x_2 + 2x_3 = 19$
$3x_1 + x_2 + 3x_3 = 15$

c) $4x_1 + 2x_2 + 6x_3 = 28$
$3x_1 + x_2 + 2x_3 = 20$
$10x_1 + 5x_2 + 15x_3 = 70$

d) $2x_1 + 4x_2 - 3x_3 = 12$
$3x_1 - 5x_2 + 2x_3 = 13$
$-x_1 + 3x_2 + 2x_3 = 17$

Solution

a) From a, we have

$$\underbrace{\begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} 8 \\ 12 \\ 5 \end{pmatrix}}_{b}$$

We need to determine the minors and the cofactors for the above matrix

Definition

A minor is a determinant of a sub matrix obtained when other elements are detected as shown below.
A cofactor is the product of (-1) i + j and a minor where
i = Ith row i = 1, 2, 3 …….
j = Jth row j = 1, 2, 3 …….

Cofactor of 4 (a_{11}) = $(-1)^{1+1} \begin{vmatrix} 3 & 1 \\ -1 & 4 \end{vmatrix} = 13$

Cofactor of -2 (a_{21}) = $(-1)^{2+1} \begin{vmatrix} 1 & -5 \\ -1 & 4 \end{vmatrix} = 1$

Cofactor of 3 (a_{31}) = $(-1)^{3+1}$ $\begin{vmatrix} 1 & -5 \\ 3 & 1 \end{vmatrix} = 16$

Cofactor of 1 (a_{12}) = $(-1)^{1+2}$ $\begin{vmatrix} -2 & 1 \\ 3 & 4 \end{vmatrix} = 11$

Cofactor of 3 (a_{22}) = $(-1)^{2+2}$ $\begin{vmatrix} 4 & -5 \\ 3 & 4 \end{vmatrix} = 31$

Cofactor of -1 (a_{23}) = $(-1)^{2+3}$ $\begin{vmatrix} 4 & 5 \\ -2 & 1 \end{vmatrix} = 6$

Cofactor of -5 (a_{13}) = $(-1)^{1+3}$ $\begin{vmatrix} -2 & 3 \\ 3 & -1 \end{vmatrix} = -7$

Cofactor of +1 (a_{23}) = $(-1)^{2+3}$ $\begin{vmatrix} 4 & 1 \\ 3 & -1 \end{vmatrix} = 7$

Cofactor of 4 (a_{33}) = $(-1)^{3+3}$ $\begin{vmatrix} 4 & 1 \\ -2 & 3 \end{vmatrix} = 14$

The matrix of C of cofactors is

$$\begin{pmatrix} 13 & 11 & -7 \\ 1 & 31 & 7 \\ 16 & 6 & 14 \end{pmatrix}$$

$$C^T = \begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix}$$ = Adjoint of the original matrix of coefficients

The original matrix of coefficients

$$= \begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix}$$

Therefore

$$\begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix} = \begin{array}{ccccc} 4 & 1 & -5 & 4 & 1 \\ -2 & 3 & 1 & -2 & 3 \\ 3 & -1 & 4 & 3 & -1 \end{array}$$

= (48 + 3 – 10) – (-45 – 4 – 8)

= 41 + 57

= 98

The inverse of the matrix of coefficients, see (*) will be

$$= \frac{1}{98}\begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix}$$

by multiplying the inverse on both sides of * we have,

$$\frac{1}{98}\begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix} \begin{pmatrix} 4 & 1 & -5 \\ -2 & 3 & 1 \\ 3 & -1 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$= \frac{1}{98}\begin{pmatrix} 13 & 1 & 16 \\ 11 & 31 & 6 \\ -7 & 7 & 14 \end{pmatrix} \begin{pmatrix} 8 \\ 12 \\ 5 \end{pmatrix}$$

$$= \frac{1}{98}\begin{pmatrix} 98 & 0 & 0 \\ 0 & 98 & 0 \\ 0 & 0 & 98 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{98}\begin{pmatrix} 196 \\ 490 \\ 98 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \\ 1 \end{pmatrix}$$

$\therefore$ $X_1 = 2, X_2 = 5, X_3 = 1$

c) $4x_1 + 2x_2 + 6x_3 = 28$
$3x_1 + x_2 + 2x_3 = 20$
$10x_1 + 5x_2 + 15x_3 = 70$

$$= \begin{pmatrix} 4 & 2 & 6 \\ 3 & 1 & 2 \\ 10 & 5 & 15 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 28 \\ 20 \\ 70 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 2 & 6 \\ 3 & 1 & 2 \\ 10 & 5 & 15 \end{pmatrix} = \begin{matrix} 4 & 2 & 6 & 4 & 2 \\ 3 & 1 & 2 & 3 & 1 \\ 10 & 5 & 15 & 10 & 5 \end{matrix}$$

$= (60 + 40 + 90) - (60 + 40 + 90)$

$= 0$

Hence the solutions of x_1, x_2, and x_3 do no exist. The equations are independent

Now work out part (b) on your own.

Cramers Rule in Solving Simultaneous Equations

Consider the following system of two linear simultaneous equations in two variables.

$a_{11} x_1 + a_{12} x_2 = b_1$ ....(i)

$a_{21} x_1 + a_{22} x_2 = b_2$ ....(ii)

after solving the equations you obtain

$$x_1 = \frac{b_1 a_{22} - b_2 a_{12}}{a_{11} a_{22} - a_{12} a_{21}} = \frac{\begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

and

$$x_2 = \frac{a_{11} b_2 - a_{21} b_1}{a_{11} a_{22} - a_{12} a_{21}} = \frac{\begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}}{\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}}$$

Solutions of x_1 and x_2 obtained this way are said to have been derived using Cramers rule, practice this method over and over to internalize it. It is advisable for exam situation since it is shorter.

Example

Solve the following systems of linear simultaneous equations by Cramers' rule:

i) $2x_1 - 5x_2 = 7$
$x_1 + 6x_2 = 9$

ii) $x_1 + 2x_2 + 4x_3 = 4$
$2x_1 + x_3 = 3$
$3x_2 + x_3 = 2$

Solutions

i. $2x_1 - 5x_2 = 7$

$x_1 + 6x_2 = 9$

can be expressed in matrix form as

$$\begin{pmatrix} 2 & -5 \\ 1 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7 \\ 9 \end{pmatrix}$$

$$\overline{A} \qquad \overline{X} \qquad \qquad \bar{b}$$

and applying cramers' rule

$$x_1 = \frac{\begin{vmatrix} 7 & -5 \\ 9 & 6 \end{vmatrix}}{\begin{vmatrix} 2 & -5 \\ 1 & 6 \end{vmatrix}} = \frac{87}{17} = 5\frac{2}{17}$$

$$x_2 = \frac{\begin{vmatrix} 2 & 7 \\ 1 & 9 \end{vmatrix}}{\begin{vmatrix} 2 & -5 \\ 1 & 6 \end{vmatrix}} = \frac{11}{17}$$

(ii) can be expressed in matrix form as

$$\begin{pmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$

and by cramers' rule

$$x_1 = \frac{\begin{vmatrix} 4 & 2 & 4 \\ 3 & 0 & 1 \\ 2 & 3 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix}} = \frac{22}{17}$$

$$x_3 = \frac{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 3 \\ 0 & 3 & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix}} = \frac{7}{17}$$

$$x_2 = \frac{\begin{vmatrix} 1 & 4 & 4 \\ 2 & 3 & 1 \\ 0 & 2 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 4 \\ 2 & 0 & 1 \\ 0 & 3 & 1 \end{vmatrix}} = \frac{9}{17}$$

Solving simultaneous Equations using matrix algebra

i. Solve the equations

2x + 3y = 13

3x + 2y = 12

in matrix format these equations can be written as

$$\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 13 \\ 12 \end{pmatrix}$$

pre multiply both sides by the inverse of the matrix

$$\Delta = \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5$$

and inverse of the matrix is

$$-\frac{1}{5}\begin{pmatrix} 2 & -3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{pmatrix}$$

Pre multiplication by inverse gives

$$\begin{pmatrix} -\frac{2}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{pmatrix} \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} -\frac{2}{5} & \frac{3}{5} \\ \frac{3}{5} & -\frac{2}{5} \end{pmatrix} \begin{pmatrix} 13 \\ 12 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Therefore x = 2 y = 3

ii. Solve the equations

4x + 2y + 3z = 4

5x + 6y + 1z = 2

2x + 3y = -1

Solution:

Writing these equations in matrix format, we get

$$A \quad \times \; B\overline{X} \; = \; \overline{b}$$

$$\begin{pmatrix} 4 & 2 & 3 \\ 5 & 6 & 1 \\ 2 & 3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix}$$

Pre-multiply both sides by the inverse

the inverse of A as found before is $A^{-1} = \begin{pmatrix} -3 & 9 & -16 \\ 2 & -6 & 11 \\ 3 & -8 & 14 \end{pmatrix}$

$$\begin{pmatrix} -3 & 9 & -16 \\ 2 & -6 & 11 \\ 3 & -8 & 14 \end{pmatrix} \begin{pmatrix} 4 & 3 & 2 \\ 5 & 6 & 1 \\ 2 & 3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -3 & 9 & -16 \\ 2 & -6 & 11 \\ 3 & -8 & 14 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 22 \\ -15 \\ -18 \end{pmatrix}$$

hence $x = 22$ $y = -15$ $z = -18$

(Note: under examination conditions it may be advisable to check the solution by substituting the value of x, y, z into any of the three original equations)